Averages

$$\text{Avg} = \frac{\text{Sum of all terms}}{\text{Total No. of terms}}$$

1. 2 dragons and 8 unicorns are bought at an average of 140 ₹, If, the average price of a unicorn is Rs.60. What is the average price of a dragon?

A) Rs.480 B) Rs.920 C) Rs.460 D) Rs.980

$\frac{2D+8U}{10} = 140$

$2D + 8\times 60 = 1400$

$2D = 1400 - 480 = 920$

$D = \frac{920}{2} = 460$

2. A family consists of two grandparents, two parents & 3 grandchildren. The average age of the grandparents is 67 years, that of the parents is 35 years and that of the grandchildren is 6 years. What is the average age of the family?

A) $28\frac{4}{9}$ B) $31\frac{5}{7}$ C) $32\frac{1}{6}$ D) $31\frac{2}{3}$

$\frac{2\times 67 + 2\times 35 + 3\times 6}{2+2+3} = \frac{222}{7}$

$7)\,222\,(31$
$\underline{21}$
12
$\underline{7}$
5

81 85 92 94 95

90

-9 $\cancel{-5}$ $+2$ $+4$ $\cancel{+5}$

$\frac{-3}{5} = -0.6$

$90 - 0.6 = 89.4$

68 74 81 83 84

75

$\cancel{-7}$ $\cancel{-1}$ $+6$ $\cancel{+8}$ $+9$

$\frac{15}{5} = +3$

$75 + 3 = 78$

Sum of the Terms

↳ avg nikalne ka shortcut (koi bich ka number lelo approximately then add or subtract from all numbers, add them and divide the result by total no. of numbers. Whatever is the result add if +ve and −if −ve to the number we assumed to get the average.

5. If the average weight of 4 men increases by 3kg when one of them weighing 90kg is replaced by another man, then the weight of the new man is?

A) 80kg B) 112kg C) 78kg D) 102kg

$\frac{4A - 90 + x}{4} = A + 3$

$\cancel{4A} - 90 + x = \cancel{4A} + 12$

$x = 12 + 90$

$= 102$

$4\times 3 = 12$

$90 + 12 = 102$

(Trick)

(new person's weight is asked)

↳ Diff = 3×4 = 12

Q = 90 + 12 ✓

90 − 12

naye bande ke add hone se weight badha to + wala lenge cuz new person's weight is asked

6. If the average weight of 4 men increases by 3kg when one of them is replaced by another man weighing 90kg, then the weight of the replaced man is?

A) 80kg B) 112kg C) 78kg D) 102kg

$\frac{4A - x + 90}{4} = A + 3$

$\cancel{4A} - x + 90 = \cancel{4A} + 12$

$x = 90 - 12$

$= 78$

(old person's weight is asked)

↳ Diff = 3×4 = 12

Q = 90 + 12

90 − 12 ✓

naye bande ke add hone se weight badha means old person's weight must be less. So −ve karenge since old person's weight is asked.

8. Ross Geller finds the average of 10 two-digit positive in... mistake, he interchanges the digits of one number, say AB. Due to this, the average becomes 1.8 less than the correct one. Find the value of |A-B|.

A) 3 B) 6 C) 2 D) 4

Correct No. $= AB = 10A+B$

Wrong No. $= BA = 10B+A$

Diff $= 1.8 \times 10 = 18$

$10A+B-(10B+A) = 18$

$10A+B-10B-A = 18$

$9A-9B = 18$

$9(A-B) = 18$

$A-B = \frac{18}{9} = 2$

10. The average wages of a worker during a fortnight comprising 15 consecutive working days was Rs.90 per day. During the first 7 days, his average wages was Rs.87/day and the average wages during the last 7 days was Rs.92 /day. What was his wage on the 8th day?

A) 83 B) 92 C) 90 D) 97

$$\overbrace{\underbrace{1\ 2\ 3\ 4\ 5\ 6\ 7}_{87},\ \underset{x}{8}\ \underbrace{9\ 10\ 11\ 12\ 13\ 14\ 15}_{92}}^{90}$$

$90 \times 15 = 87 \times 7 + x + 92 \times 7$

$x = 90 \times 15 - 87 \times 7 - 92 \times 7$

$= 97$